

VTU-ETR Seat No.

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Question Paper Version : A

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) August – 2023

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
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VTU - ETR - AUGUST - 2023

PART – I

(Research Methodology)

[50 Marks]

1. What is the significance of research?
 - a) Gathering information
 - b) Solving problems
 - c) Formulating theories
 - d) Analyzing data
2. Which of the following best defines research?
 - a) Systematic investigation
 - b) A random experiment
 - c) A personal opinion
 - d) An intuitive guess
3. Which of the following is NOT an objective of research?
 - a) Exploring new phenomena
 - b) Developing new theories
 - c) Testing pre conceived notions
 - d) Proving personal beliefs.
4. What is the difference between research methods and research methodology?
 - a) Methods are practical techniques, while methodology is the theoretical framework.
 - b) Methods focus on data analysis, while methodology focuses on data collection.
 - c) Methods are used in natural sciences, while methodology is used in social sciences.
 - d) Methods are objective, while methodology is subjective.
5. What are the criteria of good research?
 - a) Replicability, Objectivity and Reliability
 - b) Creativity, Subjectivity and Originality
 - c) Simplicity, brevity, and clarity
 - d) Validity, Complexity and comprehensiveness
6. What is the purpose of reviewing the literature in research?
 - a) To validate the researcher's hypothesis
 - b) To identify gaps in existing knowledge
 - c) To find short cuts for conducting research
 - d) To plagiarize ideas from previous studies.
7. What is the role of a theoretical framework in research?
 - a) It provides a structure for organizing the literature review.
 - b) It establishes the research objectives and hypotheses.
 - c) It guides the data collection and analysis process.
 - d) It helps in selecting the appropriate research design.
8. What is the meaning of research design?
 - a) The overall plan for conducting a research study.
 - b) The specific research method used in data collection.
 - c) The Statistical analysis technique used in research.
 - d) The process of formulating research questions.
9. Which of the following is an important principle of experimental designs?
 - a) Random sampling of participants
 - b) Use of control groups
 - c) Longitudinal data collection
 - d) Qualitative data.

10. What is the need for research design?
 - a) It helps in determining the sample size
 - b) It ensures the accuracy of data analysis.
 - c) It provides a structure for conducting research
 - d) It minimizes the ethical concerns in research.
11. What is the difference between a sample survey and a census survey?
 - a) Sample survey includes the entire population, while census survey includes a subset.
 - b) Sample survey is conducted using questionnaires while census surveys use interviews.
 - c) Sample survey is less accurate than census survey.
 - d) Sample survey is more expensive than census survey.
12. What are the types of sampling designs?
 - a) Probability sampling and non probability sampling.
 - b) Cluster sampling and convenience sampling.
 - c) Snowball sampling and purposive sampling.
 - d) Quota sampling and stratified sampling.
13. Which of the following is NOT a classification of measurement scales?
 - a) Normal Scale
 - b) Ordinal scale
 - c) Interval scale
 - d) Hierarchical scale
14. What are the sources of error in measurement?
 - a) Systematic error and random error
 - b) Measurement bias and sampling error
 - c) Observer bias and response bias
 - d) Internal validity and external validity.
15. What is the purpose of scaling in research?
 - a) To measure the central tendency of data
 - b) To determine the reliability of measurements
 - c) To assign numbers to represent characteristics
 - d) To calculate statistical significance.
16. What is the process of data preparation in research?
 - a) Collecting data from primary sources.
 - b) Analyzing data using statistical software.
 - c) Cleaning and organizing data for analysis.
 - d) Conducting data visualization techniques.
17. Which of the following is NOT a problem in the data preparation process?
 - a) Handling missing values
 - b) Dealing with outliers
 - c) Ensuring data privacy
 - d) Checking data accuracy.
18. What are the measures of central tendency?
 - a) Mean, Median and mode
 - b) Variance, standard deviation and range
 - c) Skewness, Kurtosis and correlation
 - d) Probability, confidence interval and hypothesis testing.
19. What do measures of dispersion indicate in research?
 - a) The shape of the data distribution
 - b) The spread or variability of the data
 - c) The central value of the data set
 - d) The relationship between variables.

20. What is the purpose of descriptive statistics?
a) To summarize and describe data
b) To test hypotheses and make inferences
c) To establish causal relationships
d) To predict future outcomes.
21. What is the difference between a parameter and a statistic?
a) Parameters describe populations, while statistics describe samples.
b) Parameters are qualitative, while statistics are quantitative.
c) Parameters are based on theory, while statistics are based on observation.
d) Parameters are more reliable than statistics.
22. What is the central limit theorem?
a) It states that the sample mean approaches the population mean as the sample size increases.
b) It explains the relationship between the mean and standard deviation of a sample.
c) It ensures that the sampling distribution is normally distributed.
d) It allows for the estimation of population parameters based on sample statistics.
23. What is the standard error?
a) The standard deviation of the population.
b) The standard deviation of the sample.
c) The measure of sampling error.
d) The measure of non-sampling error.
24. What is the purpose of finite population correction?
a) To adjust for sampling bias.
b) To account for non-response error.
c) To correct for measurement error.
d) To adjust the standard error for small populations.
25. What does statistical inference involve?
a) Making predictions about the future based on past data.
b) Estimating population parameters from sample statistics.
c) Analyzing the relationship between variables.
d) Describing and summarizing data.
26. If terms such as, 'positive', 'negative', 'more than', 'less than' etc are used to make a statement, then such a hypothesis is called.....
a) Relational hypotheses
b) Causal hypothesis
c) Research hypothesis
d) Directional hypothesis.
27. The tests of significance or test statistic are classified into.....
a) Parametric test
b) Non parametric test
c) Both parametric and nonparametric tests
d) None of these.
28. A two-tailed test rejects the null hypothesis if the sample mean is significantly____ than the hypothesized value of the mean of the population.
a) Higher
b) Lower
c) Higher or lower
d) None of these
29. _____ is a way to test for comparing two related samples, involving small values of 'n' that does not require the variances of the two populations to be equal.
a) Z-test
b) Paired t – test
c) f-test
d) None of these
30. As a test of independence, χ^2 test enables us to explain _____.
a) Whether or not two attributes are associated
b) The fit is considered to be a good one
c) Whether test is significant
d) None of these

31. The ANOVA technique is important in the context of all those situations where all want to compare _____
a) Two population
b) More than two populations
c) Less than two populations
d) Single population.
32. Analysis of variance in which only _____ is used to analyze the difference between more than two population means is called one way anova.
a) Two criterion (variable)
b) One criterion (variable)
c) Both two criterion (variable) and one criterion (variable)
d) None of these.
33. The basic principle of ANOVA is to test for differences among the _____ by examining the _____ within each of these samples, relative to the amount of variation between the samples.
a) Means of the populations and amount of variation
b) Estimation of the population and amount of variation
c) Means of the samples and amount of variation
d) Samples and amount of variation.
34. While applying the ANOVA techniques, the influence the uncontrolled variable is usually removed by _____
a) Multiple regression method
b) Simple linear regression method
c) One way ANOVA
d) All of these
35. What are the other assumptions in ANOVA _____
a) Various treatment groups are selected at random from the population.
b) The groups are homogeneous in variability.
c) The regression is linear and is from group to group.
d) All of these.
36. Regression analysis helps to determine _____ to measure the variability or spread of values of a dependent variable with respect to the regression line.
a) Independent value
b) Dependent value
c) Linear regression
d) Standard error of estimate.
37. Regression models are used for several purposes, including the following _____
a) Data description
b) Parameter Estimation
c) Linear regression and standard error of estimate
d) All of these.
38. Factor analysis is by for the most often used _____ of research studies, specially pertaining to social and behavioural sciences.
a) Multi functional technique
b) Multi lateral technique
c) Multi linear
d) Multivariate technique.
39. _____, in the context of factor analysis, is something like staining a microscope slide.
a) Flipping
b) Factoring
c) Rotation
d) Synchronizing.

40. In _____, the correlations are computed between pairs of respondents instead of pairs of variables.
 a) Q-type factor analysis
 b) P-type factor analysis
 c) R-type factor analysis
 d) S-type factor analysis.
41. If the outcome of an experiment is certain, that is the outcome is always the same, it is then a _____
 a) Probabilistic Experiment
 b) Quasi Experiment
 c) Deterministic Experiment
 d) Random Experiment.
42. The probability of the occurrence of a single event, such as $P(A)$ or $P(B)$, which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called _____
 a) Joint Probability
 b) A marginal probability
 c) Conditional Probability
 d) Unconditional Probability.
43. When the sample space S is countable, it is known as a _____
 a) Discrete sample space
 b) Continuous sample space
 c) Asynchronous sample space
 d) Synchronous sample space.
44. If two or more events cannot occur simultaneously in a single trail of an experiment, then such events are called _____
 a) Mutually exclusive events or disjoint events
 b) Collectively exhaustive events
 c) Both A and B
 d) Joint events.
45. The _____ is that value that occurs most often or has the greatest probability of occurring.
 a) Mean of a random variable
 b) Mode of a random variable
 c) Median of a random variable
 d) None of these
46. Two events are mutually exclusive if the occurrence of one of them prevents or rules out the occurrence of the other is called _____
 a) Collectively exhaustive events
 b) Mutually exclusive events
 c) Both A and B
 d) None of these.
47. Binomial probability distribution is a widely used probability distribution where _____
 a) There are only two possible outcomes
 b) There exists discrete random variable
 c) it's a Bernoulli process
 d) All of these
48. Mean μ and standard deviation σ are called parameters of _____
 a) Normal distribution
 b) Binomial
 c) Bernoulli Process
 d) Poisson
49. The Poisson process measures the number of occurrences of a particular outcome of a discrete random variable in a predetermined _____
 a) Time interval
 b) Space
 c) Volume
 d) All of these
50. In the _____ process, the random variable values need counting. Such a count might be number of telephone calls per hour coming into the switchboard, number of fatal traffic accidents per week in a city/state etc. and so on.
 a) Binomial distribution
 b) Normal distribution
 c) Bayesian
 d) Poisson

PART – II

Discipline Oriented Section (Computer Science and Engineering)

[50 MARKS]

51. What is the equivalent infix expression of the following postfix expression?
 $MNO + * P / Q RST / + * -$
 a) $N * (M + 0) / Q - P * (S + R / T)$
 b) $((M * (N + 0)) / P) - (Q * (R + (S / T)))$
 c) $0 * (M + N) / P - Q * (R + S / T)$
 d) $M * (N + 0) / Q - P * (R / S + T)$
52. What is the output of the following code? Void my – recursive – function (int n)

```

{
    if (n == 0)
        return ;
    printf ("%d ", n);
    my-recursive-function (n - 1);
}
Int main ( )
    my-recursive-function (10);
    return 0;
}
    
```

 a) 10
 b) 1
 c) 10 9 8 7 6 5 4 3 2 1 0
 d) 10 9 8 7 6 5 4 3 2 1
53. What is the output of following function for start pointing for first node of following linked list?
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$
 Void fun (struct node * start)

```

{
    if (start == NULL)
        return ;
    printf ("%d", start → data);
    if (start → next != NULL)
        fun (start → next → next);
    printf ("%d", start → data);
}
    
```

 a) 1 4 6 6 4 1
 b) 1 3 5 1 3 5
 c) 1 2 3 5
 d) 1 3 5 5 3 1
54. In order traversal of a tree resulted E A C K F H D B G, the pre-order traversal would return
 a) F A E K C D B H G
 b) F A E K C D H G B
 c) E A F K H D C B G
 d) F E A K D C H B G
55. Which of the following sorting procedure is the slowest?
 a) Quick sort
 b) Heap sort
 c) Shell sort
 d) Bubble sort
56. Ordered collection of objects
 a) Function
 b) Set
 c) Relation
 d) Preposition

57. What is the Cardinality of the power set of the set $\{0, 2, 3\}$
 a) 6 b) 7 c) 8 d) 9
58. Among these discrete object are
 a) Integers b) People c) Rational numbers d) All of these
59. If in a graph multiple edges between the same set of vertices are allowed, it is called
 a) Simple graphs b) Multigraphs c) Hamiltonian graph d) Euler graphs
60. Let f and g be the function from the set of integers to itself defined by $f(x) = 3x + 1$ and $g(x) = 2x + 2$. Then the composition of f and g is -----
 a) $6x + 9$ b) $6x + 3$ c) $6x + 7$ d) $6x + 8$
61. Customized products are also known as -----
 a) Bespoke products b) Generic product c) General products d) All of these
62. ----- are constricts on the services or functions offered by the system
 a) System requirements b) Functional requirements
 c) Non-functional requirement d) None of them
63. ----- helps us to design the specific test scores that we need as it shows what inputs are required and what outputs are created
 a) use case diagram b) Schema diagram c) Sequence diagram d) Activity diagram
64. ----- is best known and widely used of the agile methods.
 a) Linear programming b) Extreme programming
 c) Object oriented system development d) None of these
65. ----- occurs in real time systems use shared memory or message passing inter face
 a) Timing Errors b) Programming errors c) Integration errors d) All of these
66. Which is the following is page fault?
 a) Page fault occurs when a program accesses a page of another program
 b) Page fault occurs when a program accesses a page in main memory
 c) Page fault occurs when there is an error in particular page
 d) Page fault occurs when a program access a page which is not present in main memory
67. Which of the following is a group of bits that tells the computer to perform a particular operation?
 a) Accumulator b) Register c) Instruction code d) None of the above
68. Which of the following is performed by half adder?
 a) Binary addition operation for 2 decimal inputs
 b) Binary addition operation for 2 binary input
 c) Decimal addition operation for 2 decimal input
 d) Decimal addition operation for 2 binary inputs
69. In which of the following term the performance of cache memory is measured?
 a) Chat Ratio b) Hit Ratio c) Copy Ratio d) Data Ratio

70. Content of the program counter is added to the address part of the instruction in order to obtain the effective address is called.
 - a) Relative address mode
 - b) Index addressing mode
 - c) Register mode
 - d) Implied mode
71. The numbers of comparison required to sort the numbers 10, 20, 30, 40, 50, 60 in ascending orders using Quick sort is
 - a) 15
 - b) 25
 - c) 6
 - d) 36
72. Which of the following technique is used to solve the problems with overlapping rule problems
 - a) Greedy
 - b) Backtracking
 - c) Divide and conquer
 - d) Dynamic programming
73. The best case complexity of Insertion sort algorithm is
 - a) $\theta(n \log n)$
 - b) $\theta(n^2)$
 - c) $\theta(n)$
 - d) $\theta(\log n)$
74. The minimum numbers of moves required to move 6 disks from sources to destination is
 - a) 63
 - b) 64
 - c) 36
 - d) 31
75. The total number of solution exist to solve one Queen, Two Queens and Three Queens problems using Backtracking method are respectively
 - a) 1, 2, 3
 - b) 1, 1, 2
 - c) 1, 1, 0
 - d) 1, 0, 0
76. Which of the following is the disk space allocation method?
 - a) Contiguous Allocation
 - b) Linked Allocation
 - c) Indexed Allocation
 - d) All of the above
77. Which of the following statements true with respect to Thrashing
 - a) Is a natural consequence of virtual memory systems
 - b) Can always be avoided by swapping
 - c) Always occurs on large computers
 - d) None of above
78. Windows NT uses a ----- implementation for I/O
 - a) Message – Passing
 - b) Draft – passing
 - c) Secondary memory
 - d) Cache
79. Which one of the following is a visual (mathematical) way to determine the deadlock occurrence?
 - a) Resource allocation graph
 - b) Starvation graph
 - c) Inversion graph
 - d) None of the mentioned
80. Program always deals with :
 - a) Logical address
 - b) Absolute address
 - c) Physical address
 - d) Relative address
81. In Ethernet addressing, if all the bits are 1s, the address is
 - a) Multicast
 - b) Broadcast
 - c) Unicast
 - d) None of the above
82. Which of the following is an applications layer service?
 - a) Transferring files
 - b) Mail service
 - c) Network virtual terminal
 - d) All of the mentioned

83. Which layer divides each message into packets at the source and re-assembles them at the destination?
 a) Network layer b) Transport layer c) Data link layer d) Physical layer
84. Which types of address is 255.255.255.255 according to IPV4 addressing?
 a) Multicast address b) Limited broadcast address
 c) Loopback address d) Unicast address
85. Security features that control the access resources in the operating system
 a) Authentication b) Validation c) Identification d) Access control
86. ----- denotes the aspirations of the users and responsibility of the system to its uses
 a) UML b) OOA c) Use - cases d) Association
87. ----- identifies the same data structure and behaviour and groups them into a class
 a) Polymorphism b) Indentify c) Classification d) Inheritance
88. The protected mode is depicted in UML with the prefix of the symbol
 a) + b) - c) ~ d) #
89. The term Aggregation suits which of the following
 a) Expression a part – of – relationship and is a stronger form of an association relationship
 b) Expression a part – of – relationship and is a weaker form of an association relationship
 c) Expression an is – a relationship and is a stronger form of an association relationship
 d) Expresses an is – a – relationship and is a weaker form of an association relationship
90. ----- defines a one to many dependency between objects so that when one objects changes state, all its dependents are notified and updated automatically
 a) FACADE b) OBSERVER c) VISITOR d) ADAPTER
91. Which of the following is conflicts operation?
 a) Reads and writes from the same transaction
 b) Reads and writes from the different transactions
 c) Reads and writes from different transactions an different data items
 d) Reads and writes from different transactions on same data
92. Consider the relation $R = (A, B, C, D, E, F)$ and
 $FDS = A \rightarrow B$
 $C \rightarrow DF$
 $AC \rightarrow E$
 $D \rightarrow F$
 Decompose the FDs into 2NF
 a) $R_1(AB)$ $R_2(CDF)$ $R_3(ACE)$ b) $R_1(AB)$ $R_2(CDEF)$
 c) $R_1(ABC)$ $R_2(CDF)$ d) $R_1(AB)$ $R_2(CD)$ $R_3(EF)$
93. Which one is correct with respect to RDBMS?
 a) Primary key $\leq$ Super key $\leq$ Candidate key b) Primary key $\leq$ Candidate key $\leq$ Super key
 c) Super key $\leq$ Candidate key $\leq$ Primary key d) Super key $\leq$ Primary key $\leq$ candidate key

94. Select title
From Teacher
WHERE fees $\leq$ 200000 AND
fees $\geq$ 80,000 ;
which is the alternative query for the above
a) SELECT title
From teacher
WHERE feed BETWEEN 200000 AND 80000;
b) SELECT title
From Staff
WHERE fees BETWEEN 80,000 AND 20,0000 ;
c) SELECT title
From Staff
Where fees $\leq$ 80000 fees $\geq$ 200000
d) SELECT title from
teacher WHERE
fees BETWEEN 80000
AND 2,00000 ;
95. Sorting of relations that do not fit in memory called as
a) Internal sorting b) External sorting c) Overflow sorting d) Over load sorting
96. Customers arrive at a reception counter at an average interval of 10 minutes and the receptionist takes an average of 6 minutes for one customer. Determine the average queue length
a) 9/10 b) 7/10 c) 11/10 d) 3/10
97. Which of the following statements are not true of simulation?
a) Simulation model cannot prescribe what should be done about a problem
b) Simulation models can be used to study alternative solutions to a problem
c) Simulation models the behavior of a system
d) The equations describing the operating characteristics of the system are known
98. Simulation models can be used to obtain operating characteristics estimates in less time than with the real system using a feature of simulation called
a) Microseconds b) Warp seed c) Time compression d) None of the above
99. A multiple choice test has 30 questions there are 4 choices for each question. A student who has not studied for the test decides to answer all the questions randomly by guessing the answer for each question. Which of the following probability distributions can be used to calculate the students chance of getting at least 20 questions right?
a) Binominal Distribution b) Poisson distribution
c) Exponential distribution d) Normal distribution
100. Which of the following are techniques to perform verification of simulation model?
a) By using programming skills to write and debug the program in sub programs.
b) By tracing the intermediate results and comparing them with observed outcomes
c) By checking the simulation model output using various input combinations
d) All of the above.

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26. What is the process of data preparation in research?
- a) Collecting data from primary sources.
 - b) Analyzing data using statistical software.
 - c) Cleaning and organizing data for analysis.
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27. Which of the following is NOT a problem in the data preparation process?
- a) Handling missing values
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 - c) Ensuring data privacy
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28. What are the measures of central tendency?
- a) Mean, Median and mode
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 - c) Skewness, Kurtosis and correlation
 - d) Probability, confidence interval and hypothesis testing.
29. What do measures of dispersion indicate in research?
- a) The shape of the data distribution
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 - b) To test hypotheses and make inferences
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 - d) To predict future outcomes.

31. What is the significance of research?
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b) Solving problems
c) Formulating theories
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a) Systematic investigation
b) A random experiment
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a) Methods are practical techniques, while methodology is the theoretical framework.
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39. Which of the following is an important principle of experimental designs?
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b) Use of control groups
c) Longitudinal data collection
d) Qualitative data.
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50. As a test of independence, χ^2 test enables us to explain _____
- Whether or not two attributes are associated
 - The fit is considered to be a good one
 - Whether test is significant
 - None of these

PART – II
Discipline Oriented Section
(Computer Science and Engineering)

[50 MARKS]

51. Which of the following is conflicts operation?
a) Reads and writes from the same transaction
b) Reads and writes from the different transactions
c) Reads and writes from different transactions an different data items
d) Reads and writes from different transactions on same data
52. Consider the relation $R = (A, B, C, D, E, F)$ and
FDS = $A \rightarrow B$
 $C \rightarrow DF$
 $AC \rightarrow E$
 $D \rightarrow F$
Decompose the FDs into 2NF
a) $R_1(AB)$ $R_2(CDF)$ $R_3(ACE)$
b) $R_1(AB)$ $R_2(CDEF)$
c) $R_1(ABC)$ $R_2(CDF)$
d) $R_1(AB)$ $R_2(CD)$ $R_3(EF)$
53. Which one is correct with respect to RDBMS?
a) Primary key $\leq$ Super key $\leq$ Candidate key
b) Primary key $\leq$ Candidate key $\leq$ Super key
c) Super key $\leq$ Candidate key $\leq$ Primary key
d) Super key $\leq$ Primary key $\leq$ candidate key
54. Select title
From Teacher
WHERE fees $\leq$ 200000 AND
fees $\geq$ 80,000 ;
which is the alternative query for the above
a) SELECT title
From teacher
WHERE feed BETWEEN 200000 AND 80000;
b) SELECT title
From Staff
WHERE fees BETWEEN 80,000 AND 20,0000 ;
c) SELECT title
From Staff
Where fees $\leq$ 80000 fees $\geq$ 200000
d) SELECT title from
teacher WHERE
fees BETWEEN 80000
AND 2,00000 ;
55. Sorting of relations that do not fit in memory called as
a) Internal sorting
b) External sorting
c) Overflow sorting
d) Over load sorting

56. Customers arrive at a reception counter at an average interval of 10 minutes and the receptionist takes an average of 6 minutes for one customer. Determine the average queue length
 a) 9/10 b) 7/10 c) 11/10 d) 3/10
57. Which of the following statements are not true of simulation?
 a) Simulation model cannot prescribe what should be done about a problem
 b) Simulation models can be used to study alternative solutions to a problem
 c) Simulation models the behavior of a system
 d) The equations describing the operating characteristics of the system are known
58. Simulation models can be used to obtain operating characteristics estimates in less time than with the real system using a feature of simulation called
 a) Microseconds b) Warp seed
 c) Time compression d) None of the above
59. A multiple choice test has 30 questions there are 4 choices for each question. A student who has not studied for the test decides to answer all the questions randomly by guessing the answer for each question. Which of the following probability distributions can be used to calculate the students chance of getting at least 20 questions right?
 a) Binominal Distribution b) Poisson distribution
 c) Exponential distribution d) Normal distribution
60. Which of the following are techniques to perform verification of simulation model?
 a) By using programming skills to write and debug the program in sub programs.
 b) By tracing the intermediate results and comparing them with observed outcomes
 c) By checking the simulation model output using various input combinations
 d) All of the above.
61. In Ethernet addressing, if all the bits are 1s, the address is
 a) Multicast b) Broadcast c) Unicast d) None of the above
62. Which of the following is an applications layer service?
 a) Transferring files b) Mail service
 c) Network virtual terminal d) All of the mentioned
63. Which layer divides each message into packets at the source and re-assembles them at the destination?
 a) Network layer b) Transport layer c) Data link layer d) Physical layer
64. Which types of address is 255.255.255.255 according to IPV4 addressing?
 a) Multicast address b) Limited broadcast address
 c) Loopback address d) Unicast address
65. Security features that control the access resources in the operating system
 a) Authentication b) Validation c) Identification d) Access control
66. ----- denotes the aspirations of the users and responsibility of the system to its uses
 a) UML b) OOA c) Use - cases d) Association
67. ----- identifies the same data structure and behaviour and groups them into a class
 a) Polymorphism b) Identify c) Classification d) Inheritance

68. The protected mode is depicted in UML with the prefix of the symbol
a) + b) – c) ~ d) #
69. The term Aggregation suits which of the following
a) Expression a part – of – relationship and is a stronger form of an association relationship
b) Expression a part – of – relationship and is a weaker form of an association relationship
c) Expression an is – a relationship and is a stronger form of an association relationship
d) Expresses an is – a – relationship and is a weaker form of an association relationship
70. ----- defines a one to many dependency between objects so that when one objects changes state, all its dependents are notified and updated automatically
a) FACADE b) OBSERVER c) VISITOR d) ADAPTER
71. Customized products are also known as -----
a) Bespoke products b) Generic product c) General products d) All of these
72. ----- are constricts on the services or functions offered by the system
a) System requirements b) Functional requirements
c) Non-functional requirement d) None of them
73. ----- helps us to design the specific test scores that we need as it shows what inputs are required and what outputs are created
a) use case diagram b) Schema diagram c) Sequence diagram d) Activity diagram
74. ----- is best known and widely used of the agile methods.
a) Linear programming b) Extreme programming
c) Object oriented system development d) None of these
75. ----- occurs in real time systems use shared memory or message passing inter face
a) Timing Errors b) Programming errors c) Integration errors d) All of these
76. Which is the following is page fault?
a) Page fault occurs when a program accesses a page of another program
b) Page fault occurs when a program accesses a page in main memory
c) Page fault occurs when there is an error in particular page
d) Page fault occurs when a program access a page which is not present in main memory
77. Which of the following is a group of bits that tells the computer to perform a particular operation?
a) Accumulator b) Register c) Instruction code d) None of the above
78. Which of the following is performed by half adder?
a) Binary addition operation for 2 decimal inputs
b) Binary addition operation for 2 binary input
c) Decimal addition operation for 2 decimal input
d) Decimal addition operation for 2 binary inputs
79. In which of the following term the performance of cache memory is measured?
a) Chat Ratio b) Hit Ratio c) Copy Ratio d) Data Ratio

80. Content of the program counter is added to the address part of the instruction in order to obtain the effective address is called.
- Relative address mode
 - Index addressing mode
 - Register mode
 - Implied mode
81. What is the equivalent infix expression of the following postfix expression?
 $MNO + * P / Q RST / + * -$
- $N * (M + 0) / Q - P * (S + R / T)$
 - $(((M * (N + 0)) / P) - (Q * (R + (S / T))))$
 - $0 * (M + N) / P - Q * (R + S / T)$
 - $M * (N + 0) / Q - P * (R / S + T)$
82. What is the output of the following code? Void my – recursive – function (int n)
- ```

{
 if (n == 0)
 return ;
 printf("%d ", n);
 my-recursive-function (n - 1);
}
Int main ()
 my-recursive-function (10);
 return 0;
}

```
- 10
  - 1
  - 10 9 8 7 6 5 4 3 2 1 0
  - 10 9 8 7 6 5 4 3 2 1
83. What is the output of following function for start pointing for first node of following linked list?  
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$   
 Void fun (struct node \* start)
- ```

{
    if (start == NULL)
        return ;
    printf("%d", start -> data);
    if (start -> next != NULL)
        fun (start -> next -> next);
    printf((" %d", start -> data);
}
    
```
- 1 4 6 6 4 1
 - 1 3 5 1 3 5
 - 1 2 3 5
 - 1 3 5 5 3 1
84. In order traversal of a tree resulted E A C K F H D B G, the pre-order traversal would return
- F A E K C D B H G
 - F A E K C D H G B
 - E A F K H D C B G
 - F E A K D C H B G
85. Which of the following sorting procedure is the slowest?
- Quick sort
 - Heap sort
 - Shell sort
 - Bubble sort
86. Ordered collection of objects
- Function
 - Set
 - Relation
 - Preposition
87. What is the Cardinality of the power set of the set $\{0, 2, 3\}$
- 6
 - 7
 - 8
 - 9

88. Among these discrete object are
a) Integers b) People c) Rational numbers d) All of these
89. If in a graph multiple edges between the same set of vertices are allowed, it is called
a) Simple graphs b) Multigraphs c) Hamiltonian graph d) Euler graphs
90. Let f and g be the function from the set of integers to itself defined by $f(x) = 3x + 1$ and $g(x) = 2x + 2$. Then the composition of f and g is -----.
a) $6x + 9$ b) $6x + 3$ c) $6x + 7$ d) $6x + 8$
91. The numbers of comparison required to sort the numbers 10, 20, 30, 40, 50, 60 in ascending orders using Quick sort is
a) 15 b) 25 c) 6 d) 36
92. Which of the following technique is used to solve the problems with overlapping rule problems
a) Greedy b) Backtracking
c) Divide and conquer d) Dynamic programming
93. The best case complexity of Insertion sort algorithm is
a) $\theta(n \log n)$ b) $\theta(n^2)$ c) $\theta(n)$ d) $\theta(\log n)$
94. The minimum numbers of moves required to move 6 disks from sources to destination is
a) 63 b) 64 c) 36 d) 31
95. The total number of solution exist to solve one Queen, Two Queens and Three Queens problems using Backtracking method are respectively
a) 1, 2, 3 b) 1, 1, 2 c) 1, 1, 0 d) 1, 0, 0
96. Which of the following is the disk space allocation method?
a) Contiguous Allocation b) Linked Allocation
c) Indexed Allocation d) All of the above
97. Which of the following statements true with respect to Thrashing
a) Is a natural consequence of virtual memory systems
b) Can always be avoided by swapping
c) Always occurs on large computers
d) None of above
98. Windows NT uses a ----- implementation for I/O
a) Message – Passing b) Draft – passing c) Secondary memory d) Cache
99. Which one of the following is a visual (mathematical) way to determine the deadlock occurrence?
a) Resource allocation graph b) Starvation graph
c) Inversion graph d) None of the mentioned
100. Program always deals with :
a) Logical address b) Absolute address c) Physical address d) Relative address

* * * * *

VTU-ETR Seat No.

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Question Paper Version : C

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) August – 2023

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

VTU - ETR - AUGUST - 2023

PART – I

(Research Methodology)

[50 Marks]

1. What is the difference between a parameter and a statistic?
 - a) Parameters describe populations, while statistics describe samples.
 - b) Parameters are qualitative, while statistics are quantitative.
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5. What does statistical inference involve?
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6. If terms such as, 'positive', 'negative', 'more than', 'less than' etc are used to make a statement, then such a hypothesis is called.....
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7. The tests of significance or test statistic are classified into.....
 - a) Parametric test
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8. A two-tailed test rejects the null hypothesis if the sample mean is significantly_____ than the hypothesized value of the mean of the population.
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9. _____ is a way to test for comparing two related samples, involving small values of 'n' that does not require the variances of the two populations to be equal.
 - a) Z-test
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10. As a test of independence, X^2 test enables us to explain _____
a) Whether or not two attributes are associated b) The fit is considered to be a good one
c) Whether test is significant d) None of these
11. If the outcome of an experiment is certain, that is the outcome is always the same, it is then a _____
a) Probabilistic Experiment b) Quasi Experiment
c) Deterministic Experiment d) Random Experiment.
12. The probability of the occurrence of a single event, such as $P(A)$ or $P(B)$, which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called _____
a) Joint Probability b) A marginal probability
c) Conditional Probability d) Unconditional Probability.
13. When the sample space S is countable, it is known as a _____
a) Discrete sample space b) Continuous sample space
c) Asynchronous sample space d) Synchronous sample space.
14. If two or more events cannot occur simultaneously in a single trail of an experiment, then such events are called _____
a) Mutually exclusive events or disjoint events b) Collectively exhaustive events
c) Both A and B d) Joint events.
15. The _____ is that value that occurs most often or has the greatest probability of occurring.
a) Mean of a random variable b) Mode of a random variable
c) Median of a random variable d) None of these
16. Two events are mutually exclusive if the occurrence of one of them prevents or rules out the occurrence of the other is called _____
a) Collectively exhaustive events b) Mutually exclusive events
c) Both A and B d) None of these.
17. Binomial probability distribution is a widely used probability distribution where _____
a) There are only two possible outcomes b) There exists discrete random variable
c) it's a Bernoulli process d) All of these
18. Mean μ and standard deviation σ are called parameters of _____
a) Normal distribution b) Binomial
c) Bernoulli Process d) Poisson
19. The Poisson process measures the number of occurrences of a particular outcome of a discrete random variable in a predetermined _____
a) Time interval b) Space
c) Volume d) All of these
20. In the _____ process, the random variable values need counting. Such a count might be number of telephone calls per hour coming into the switchboard, number of fatal traffic accidents per week in a city/state etc. and so on.
a) Binomial distribution b) Normal distribution
c) Bayesian d) Poisson

21. The ANOVA technique is important in the context of all those situations where all want to compare _____.
a) Two population
b) More than two populations
c) Less than two populations
d) Single population.
22. Analysis of variance in which only _____ is used to analyze the difference between more than two population means is called one way anova.
a) Two criterion (variable)
b) One criterion (variable)
c) Both two criterion (variable) and one criterion (variable)
d) None of these.
23. The basic principle of ANOVA is to test for differences among the _____ by examining the _____ within each of these samples, relative to the amount of variation between the samples.
a) Means of the populations and amount of variation
b) Estimation of the population and amount of variation
c) Means of the samples and amount of variation
d) Samples and amount of variation.
24. While applying the ANOVA techniques, the influence the uncontrolled variable is usually removed by _____.
a) Multiple regression method
b) Simple linear regression method
c) One way ANOVA
d) All of these
25. What are the other assumptions in ANOVA _____.
a) Various treatment groups are selected at random from the population.
b) The groups are homogeneous in variability.
c) The regression is linear and is from group to group.
d) All of these.
26. Regression analysis helps to determine _____ to measure the variability or spread of values of a dependent variable with respect to the regression line.
a) Independent value
b) Dependent value
c) Linear regression
d) Standard error of estimate.
27. Regression models are used for several purposes, including the following _____.
a) Data description
b) Parameter Estimation
c) Linear regression and standard error of estimate
d) All of these.
28. Factor analysis is by for the most often used _____ of research studies, specially pertaining to social and behavioural sciences.
a) Multi functional technique
b) Multi lateral technique
c) Multi linear
d) Multivariate technique.
29. _____, in the context of factor analysis, is something like staining a microscope slide.
a) Flipping
b) Factoring
c) Rotation
d) Synchronizing.
30. In _____, the correlations are computed between pairs of respondents instead of pairs of variables.
a) Q-type factor analysis
b) P-type factor analysis
c) R-type factor analysis
d) S-type factor analysis.

31. What is the difference between a sample survey and a census survey?
a) Sample survey includes the entire population, while census survey includes a subset.
b) Sample survey is conducted using questionnaires while census surveys use interviews.
c) Sample survey is less accurate than census survey.
d) Sample survey is more expensive than census survey.
32. What are the types of sampling designs?
a) Probability sampling and non probability sampling.
b) Cluster sampling and convenience sampling.
c) Snowball sampling and purposive sampling.
d) Quota sampling and stratified sampling.
33. Which of the following is NOT a classification of measurement scales?
a) Normal Scale b) Ordinal scale c) Interval scale d) Hierarchical scale
34. What are the sources of error in measurement?
a) Systematic error and random error b) Measurement bias and sampling error
c) Observer bias and response bias d) Internal validity and external validity.
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55. The total number of solution exist to solve one Queen, Two Queens and Three Queens problems using Backtracking method are respectively
a) 1, 2, 3 b) 1, 1, 2 c) 1, 1, 0 d) 1, 0, 0
56. Which of the following is the disk space allocation method?
a) Contiguous Allocation b) Linked Allocation
c) Indexed Allocation d) All of the above
57. Which of the following statements true with respect to Thrashing
a) Is a natural consequence of virtual memory systems
b) Can always be avoided by swapping
c) Always occurs on large computers
d) None of above
58. Windows NT uses a ----- implementation for I/O
a) Message – Passing b) Draft – passing c) Secondary memory d) Cache
59. Which one of the following is a visual (mathematical) way to determine the deadlock occurrence?
a) Resource allocation graph b) Starvation graph
c) Inversion graph d) None of the mentioned
60. Program always deals with :
a) Logical address b) Absolute address c) Physical address d) Relative address
61. Which of the following is conflicts operation?
a) Reads and writes from the same transaction
b) Reads and writes from the different transactions
c) Reads and writes from different transactions an different data items
d) Reads and writes from different transactions on same data

Ver-C : CS/IS 6 of 10

62. Consider the relation $R = (A, B, C, D, E, F)$ and
 FDS = $A \rightarrow B$
 $C \rightarrow DF$
 $AC \rightarrow E$
 $D \rightarrow F$
 Decompose the FDs into 2NF
 a) $R_1(AB) \quad R_2(CDF) \quad R_3(ACE)$
 c) $R_1(ABC) \quad R_2(CDF)$
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63. Which one is correct with respect to RDBMS?
 a) Primary key $\leq$ Super key $\leq$ Candidate key
 c) Super key $\leq$ Candidate key $\leq$ Primary key
 b) Primary key $\leq$ Candidate key $\leq$ Super key
 d) Super key $\leq$ Primary key $\leq$ candidate key
64. Select title
 From Teacher
 WHERE fees $\leq$ 200000 AND
 fees $\geq$ 80,000 ;
 which is the alternative query for the above
 a) SELECT title
 From teacher
 WHERE feed BETWEEN 200000 AND 80000;
 b) SELECT title
 From Staff
 WHERE fees BETWEEN 80,000 AND 20,0000 ;
 c) SELECT title
 From Staff
 Where fees $\leq$ 80000 fees $\geq$ 200000
 d) SELECT title from
 teacher WHERE
 fees BETWEEN 80000
 AND 2,00000 ;
65. Sorting of relations that do not fit in memory called as
 a) Internal sorting b) External sorting c) Overflow sorting d) Over load sorting
66. Customers arrive at a reception counter at an average interval of 10 minutes and the receptionist takes an average of 6 minutes for one customer. Determine the average queue length
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68. Simulation models can be used to obtain operating characteristics estimates in less time than with the real system using a feature of simulation called
 a) Microseconds b) Warp seed c) Time compression d) None of the above

69. A multiple choice test has 30 questions there are 4 choices for each question. A student who has not studied for the test decides to answer all the questions randomly by guessing the answer for each question. Which of the following probability distributions can be used to calculate the students chance of getting at least 20 questions right?
- a) Binominal Distribution
 - b) Poisson distribution
 - c) Exponential distribution
 - d) Normal distribution
70. Which of the following are techniques to perform verification of simulation model?
- a) By using programming skills to write and debug the program in sub programs.
 - b) By tracing the intermediate results and comparing them with observed outcomes
 - c) By checking the simulation model output using various input combinations
 - d) All of the above.
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- a) Multicast
 - b) Broadcast
 - c) Unicast
 - d) None of the above
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- a) Transferring files
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74. Which types of address is 255.255.255.255 according to IPV4 addressing?
- a) Multicast address
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 - c) Loopback address
 - d) Unicast address
75. Security features that control the access resources in the operating system
- a) Authentication
 - b) Validation
 - c) Identification
 - d) Access control
76. ----- denotes the aspirations of the users and responsibility of the system to its uses
- a) UML
 - b) OOA
 - c) Use - cases
 - d) Association
77. ----- identifies the same data structure and behaviour and groups them into a class
- a) Polymorphism
 - b) Identify
 - c) Classification
 - d) Inheritance
78. The protected mode is depicted in UML with the prefix of the symbol
- a) +
 - b) -
 - c) ~
 - d) #
79. The term Aggregation suits which of the following
- a) Expression a part – of – relationship and is a stronger form of an association relationship
 - b) Expression a part – of – relationship and is a weaker form of an association relationship
 - c) Expression an is – a relationship and is a stronger form of an association relationship
 - d) Expresses an is – a – relationship and is a weaker form of an association relationship
80. ----- defines a one to many dependency between objects so that when one objects changes state, all its dependents are notified and updated automatically
- a) FACADE
 - b) OBSERVER
 - c) VISITOR
 - d) ADAPTER

81. Customized products are also known as -----
 a) Bespoke products b) Generic product c) General products d) All of these
82. ----- are constricts on the services or functions offered by the system
 a) System requirements b) Functional requirements
 c) Non-functional requirement d) None of them
83. ----- helps us to design the specific test scores that we need as it shows what inputs are required and what outputs are created
 a) use case diagram b) Schema diagram c) Sequence diagram d) Activity diagram
84. ----- is best known and widely used of the agile methods.
 a) Linear programming b) Extreme programming
 c) Object oriented system development d) None of these
85. ----- occurs in real time systems use shared memory or message passing inter face
 a) Timing Errors b) Programming errors c) Integration errors d) All of these
86. Which is the following is page fault?
 a) Page fault occurs when a program accesses a page of another program
 b) Page fault occurs when a program accesses a page in main memory
 c) Page fault occurs when there is an error in particular page
 d) Page fault occurs when a program access a page which is not present in main memory
87. Which of the following is a group of bits that tells the computer to perform a particular operation?
 a) Accumulator b) Register c) Instruction code d) None of the above
88. Which of the following is performed by half adder?
 a) Binary addition operation for 2 decimal inputs
 b) Binary addition operation for 2 binary input
 c) Decimal addition operation for 2 decimal input
 d) Decimal addition operation for 2 binary inputs
89. In which of the following term the performance of cache memory is measured?
 a) Chat Ratio b) Hit Ratio c) Copy Ratio d) Data Ratio
90. Content of the program counter is added to the address part of the instruction in order to obtain the effective address is called.
 a) Relative address mode b) Index addressing mode
 c) Register mode d) Implied mode
91. What is the equivalent infix expression of the following postfix expression?
 $MNO + * P / QRST / + * -$
 a) $N * (M + 0) / Q - P * (S + R / T)$
 b) $((M * (N + 0)) / P) - (Q * (R + (S / T)))$
 c) $0 * (M + N) / P - Q * (R + S / T)$
 d) $M * (N + 0) / Q - P * (R / S + T)$

92. What is the output of the following code? Void my – recursive – function (int n)
- ```

{
 if (n == 0)
 return ;
 printf(“%d ”, n) ;
 my-recursive-function (n – 1) ;
}
Int main ()
 my-recursive-function (10) ;
 return 0 ;
}

```
- a) 10  
b) 1  
c) 10 9 8 7 6 5 4 3 2 1 0  
d) 10 9 8 7 6 5 4 3 2 1
93. What is the output of following function for start pointing for first node of following linked list?
- 1 → 2 → 3 → 4 → 5 → 6
- ```

Void fun (struct node * start)
{
    if (start == NULL)
        return ;
    printf(“%d”, start → data) ;
    if (start → next != NULL)
        fun (start → next → next) ;
    printf(“%d”, start → data) ;
}

```
- a) 1 4 6 6 4 1
b) 1 3 5 1 3 5
c) 1 2 3 5
d) 1 3 5 5 3 1
94. In order traversal of a tree resulted E A C K F H D B G, the pre-order traversal would return
- a) F A E K C D B H G
b) F A E K C D H G B
c) E A F K H D C B G
d) F E A K D C H B G
95. Which of the following sorting procedure is the slowest?
- a) Quick sort
b) Heap sort
c) Shell sort
d) Bubble sort
96. Ordered collection of objects
- a) Function
b) Set
c) Relation
d) Preposition
97. What is the Cardinality of the power set of the set {0, 2, 3}
- a) 6
b) 7
c) 8
d) 9
98. Among these discrete object are
- a) Integers
b) People
c) Rational numbers
d) All of these
99. If in a graph multiple edges between the same set of vertices are allowed, it is called
- a) Simple graphs
b) Multigraphs
c) Hamiltonian graph
d) Euler graphs
100. Let f and g be the function from the set of integers to itself defined by $f(x) = 3x + 1$ and $g(x) = 2x + 2$. Then the composition of f and g is -----.
- a) $6x + 9$
b) $6x + 3$
c) $6x + 7$
d) $6x + 8$

* * * * *

VTU-ETR Seat No.

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Question Paper Version : D

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) August – 2023

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

VTU - ETR - AUGUST - 2023

PART – I

(Research Methodology)

[50 Marks]

1. The ANOVA technique is important in the context of all those situations where all want to compare _____.
a) Two population
b) More than two populations
c) Less than two populations
d) Single population.
2. Analysis of variance in which only _____ is used to analyze the difference between more than two population means is called one way anova.
a) Two criterion (variable)
b) One criterion (variable)
c) Both two criterion (variable) and one criterion (variable)
d) None of these.
3. The basic principle of ANOVA is to test for differences among the _____ by examining the _____ within each of these samples, relative to the amount of variation between the samples.
a) Means of the populations and amount of variation
b) Estimation of the population and amount of variation
c) Means of the samples and amount of variation
d) Samples and amount of variation.
4. While applying the ANOVA techniques, the influence the uncontrolled variable is usually removed by _____.
a) Multiple regression method
b) Simple linear regression method
c) One way ANOVA
d) All of these
5. What are the other assumptions in ANOVA _____.
a) Various treatment groups are selected at random from the population.
b) The groups are homogeneous in variability.
c) The regression is linear and is from group to group.
d) All of these.
6. Regression analysis helps to determine _____ to measure the variability or spread of values of a dependent variable with respect to the regression line.
a) Independent value
b) Dependent value
c) Linear regression
d) Standard error of estimate.
7. Regression models are used for several purposes, including the following _____.
a) Data description
b) Parameter Estimation
c) Linear regression and standard error of estimate
d) All of these.
8. Factor analysis is by for the most often used _____ of research studies, specially pertaining to social and behavioural sciences.
a) Multi functional technique
b) Multi lateral technique
c) Multi linear
d) Multivariate technique.

9. _____, in the context of factor analysis, is something like staining a microscope slide.
a) Flipping
b) Factoring
c) Rotation
d) Synchronizing.
10. In _____, the correlations are computed between pairs of respondents instead of pairs of variables.
a) Q-type factor analysis
b) P-type factor analysis
c) R-type factor analysis
d) S-type factor analysis.
11. What is the significance of research?
a) Gathering information
b) Solving problems
c) Formulating theories
d) Analyzing data
12. Which of the following best defines research?
a) Systematic investigation
b) A random experiment
c) A personal opinion
d) An intuitive guess
13. Which of the following is NOT an objective of research?
a) Exploring new phenomena
b) Developing new theories
c) Testing pre conceived notions
d) Proving personal beliefs.
14. What is the difference between research methods and research methodology?
a) Methods are practical techniques, while methodology is the theoretical framework.
b) Methods focus on data analysis, while methodology focuses on data collection.
c) Methods are used in natural sciences, while methodology is used in social sciences.
d) Methods are objective, while methodology is subjective.
15. What are the criteria of good research?
a) Replicability, Objectivity and Reliability
b) Creativity, Subjectivity and Originality
c) Simplicity, brevity, and clarity
d) Validity, Complexity and comprehensiveness
16. What is the purpose of reviewing the literature in research?
a) To validate the researcher's hypothesis
b) To identify gaps in existing knowledge
c) To find short cuts for conducting research
d) To plagiarize ideas from previous studies.
17. What is the role of a theoretical framework in research?
a) It provides a structure for organizing the literature review.
b) It establishes the research objectives and hypotheses.
c) It guides the data collection and analysis process.
d) It helps in selecting the appropriate research design.
18. What is the meaning of research design?
a) The overall plan for conducting a research study.
b) The specific research method used in data collection.
c) The Statistical analysis technique used in research.
d) The process of formulating research questions.
19. Which of the following is an important principle of experimental designs?
a) Random sampling of participants
b) Use of control groups
c) Longitudinal data collection
d) Qualitative data.

20. What is the need for research design?
a) It helps in determining the sample size
b) It ensures the accuracy of data analysis.
c) It provides a structure for conducting research
d) It minimizes the ethical concerns in research.
21. If the outcome of an experiment is certain, that is the outcome is always the same, it is then a _____
a) Probabilistic Experiment b) Quasi Experiment
c) Deterministic Experiment d) Random Experiment.
22. The probability of the occurrence of a single event, such as $P(A)$ or $P(B)$, which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called _____
a) Joint Probability b) A marginal probability
c) Conditional Probability d) Unconditional Probability.
23. When the sample space S is countable, it is known as a _____
a) Discrete sample space b) Continuous sample space
c) Asynchronous sample space d) Synchronous sample space.
24. If two or more events cannot occur simultaneously in a single trail of an experiment, then such events are called _____
a) Mutually exclusive events or disjoint events b) Collectively exhaustive events
c) Both A and B d) Joint events.
25. The _____ is that value that occurs most often or has the greatest probability of occurring.
a) Mean of a random variable b) Mode of a random variable
c) Median of a random variable d) None of these
26. Two events are mutually exclusive if the occurrence of one of them prevents or rules out the occurrence of the other is called _____
a) Collectively exhaustive events b) Mutually exclusive events
c) Both A and B d) None of these.
27. Binomial probability distribution is a widely used probability distribution where _____
a) There are only two possible outcomes b) There exists discrete random variable
c) it's a Bernoulli process d) All of these
28. Mean μ and standard deviation σ are called parameters of _____
a) Normal distribution b) Binomial
c) Bernoulli Process d) Poisson
29. The Poisson process measures the number of occurrences of a particular outcome of a discrete random variable in a predetermined _____
a) Time interval b) Space c) Volume d) All of these
30. In the _____ process, the random variable values need counting. Such a count might be number of telephone calls per hour coming into the switchboard, number of fatal traffic accidents per week in a city/state etc. and so on.
a) Binomial distribution b) Normal distribution
c) Bayesian d) Poisson

31. What is the difference between a parameter and a statistic?
- Parameters describe populations, while statistics describe samples.
 - Parameters are qualitative, while statistics are quantitative.
 - Parameters are based on theory, while statistics are based on observation.
 - Parameters are more reliable than statistics.
32. What is the central limit theorem?
- It states that the sample mean approaches the population mean as the sample size increases.
 - It explains the relationship between the mean and standard deviation of a sample.
 - It ensures that the sampling distribution is normally distributed.
 - It allows for the estimation of population parameters based on sample statistics.
33. What is the standard error?
- The standard deviation of the population.
 - The standard deviation of the sample.
 - The measure of sampling error.
 - The measure of non-sampling error.
34. What is the purpose of finite population correction?
- To adjust for sampling bias.
 - To account for non-response error.
 - To correct for measurement error.
 - To adjust the standard error for small populations.
35. What does statistical inference involve?
- Making predictions about the future based on past data.
 - Estimating population parameters from sample statistics.
 - Analyzing the relationship between variables.
 - Describing and summarizing data.
36. If terms such as, 'positive', 'negative', 'more than', 'less than' etc are used to make a statement, then such a hypothesis is called.....
- Relational hypotheses
 - Causal hypothesis
 - Research hypothesis
 - Directional hypothesis.
37. The tests of significance or test statistic are classified into.....
- Parametric test
 - Non parametric test
 - Both parametric and nonparametric tests
 - None of these.
38. A two-tailed test rejects the null hypothesis if the sample mean is significantly____ than the hypothesized value of the mean of the population.
- Higher
 - Lower
 - Higher or lower
 - None of these
39. _____ is a way to test for comparing two related samples, involving small values of 'n' that does not require the variances of the two populations to be equal.
- Z-test
 - Paired t – test
 - f-test
 - None of these
40. As a test of independence, X^2 test enables us to explain _____
- Whether or not two attributes are associated
 - The fit is considered to be a good one
 - Whether test is significant
 - None of these

41. What is the difference between a sample survey and a census survey?
a) Sample survey includes the entire population, while census survey includes a subset.
b) Sample survey is conducted using questionnaires while census surveys use interviews.
c) Sample survey is less accurate than census survey.
d) Sample survey is more expensive than census survey.
42. What are the types of sampling designs?
a) Probability sampling and non probability sampling.
b) Cluster sampling and convenience sampling.
c) Snowball sampling and purposive sampling.
d) Quota sampling and stratified sampling.
43. Which of the following is NOT a classification of measurement scales?
a) Normal Scale b) Ordinal scale c) Interval scale d) Hierarchical scale
44. What are the sources of error in measurement?
a) Systematic error and random error b) Measurement bias and sampling error
c) Observer bias and response bias d) Internal validity and external validity.
45. What is the purpose of scaling in research?
a) To measure the central tendency of data
b) To determine the reliability of measurements
c) To assign numbers to represent characteristics
d) To calculate statistical significance.
46. What is the process of data preparation in research?
a) Collecting data from primary sources. b) Analyzing data using statistical software.
c) Cleaning and organizing data for analysis. d) Conducting data visualization techniques.
47. Which of the following is NOT a problem in the data preparation process?
a) Handling missing values b) Dealing with outliers
c) Ensuring data privacy d) Checking data accuracy.
48. What are the measures of central tendency?
a) Mean, Median and mode
b) Variance, standard deviation and range
c) Skewness, Kurtosis and correlation
d) Probability, confidence interval and hypothesis testing.
49. What do measures of dispersion indicate in research?
a) The shape of the data distribution b) The spread or variability of the data
c) The central value of the data set d) The relationship between variables.
50. What is the purpose of descriptive statistics?
a) To summarize and describe data b) To test hypotheses and make inferences
c) To establish causal relationships d) To predict future outcomes.

PART – II

Discipline Oriented Section (Computer Science and Engineering)

[50 MARKS]

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a) Multicast b) Broadcast c) Unicast d) None of the above
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c) $0 * (M + N) / P - Q * (R + S / T)$ d) $M * (N + O) / Q - P * (R / S + T)$

a) 10 b) 1
c) 10 9 8 7 6 5 4 3 2 1 0 d) 10 9 8 7 6 5 4 3 2 1

63. What is the output of following function for start pointing for first node of following linked list?

a) 1 4 6 6 4 1 b) 1 3 5 1 3 5 c) 1 2 3 5 d) 1 3 5 5 3 1

64. In order traversal of a tree resulted E A C K F H D B G, the pre-order traversal would return

a) F A E K C D B H G
b) F A E K C D H G B
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65. Which of the following sorting procedure is the slowest?

a) Quick sort b) Heap sort c) Shell sort d) Bubble sort

66. Ordered collection of objects

a) Function b) Set c) Relation d) Preposition

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 - d) Normal distribution
80. Which of the following are techniques to perform verification of simulation model?
- a) By using programming skills to write and debug the program in sub programs.
 - b) By tracing the intermediate results and comparing them with observed outcomes
 - c) By checking the simulation model output using various input combinations
 - d) All of the above.
81. The numbers of comparison required to sort the numbers 10, 20, 30, 40, 50, 60 in ascending orders using Quick sort is
- a) 15
 - b) 25
 - c) 6
 - d) 36
82. Which of the following technique is used to solve the problems with overlapping rule problems
- a) Greedy
 - b) Backtracking
 - c) Divide and conquer
 - d) Dynamic programming
83. The best case complexity of Insertion sort algorithm is
- a) $\theta(n \log n)$
 - b) $\theta(n^2)$
 - c) $\theta(n)$
 - d) $\theta(\log n)$
84. The minimum numbers of moves required to move 6 disks from sources to destination is
- a) 63
 - b) 64
 - c) 36
 - d) 31
85. The total number of solution exist to solve one Queen, Two Queens and Three Queens problems using Backtracking method are respectively
- a) 1, 2, 3
 - b) 1, 1, 2
 - c) 1, 1, 0
 - d) 1, 0, 0
86. Which of the following is the disk space allocation method?
- a) Contiguous Allocation
 - b) Linked Allocation
 - c) Indexed Allocation
 - d) All of the above
87. Which of the following statements true with respect to Thrashing
- a) Is a natural consequence of virtual memory systems
 - b) Can always be avoided by swapping
 - c) Always occurs on large computers
 - d) None of above
88. Windows NT uses a ----- implementation for I/O
- a) Message – Passing
 - b) Draft – passing
 - c) Secondary memory
 - d) Cache

89. Which one of the following is a visual (mathematical) way to determine the deadlock occurrence?
a) Resource allocation graph b) Starvation graph
c) Inversion graph d) None of the mentioned
90. Program always deals with :
a) Logical address b) Absolute address c) Physical address d) Relative address
91. Customized products are also known as -----
a) Bespoke products b) Generic product c) General products d) All of these
92. ----- are constricts on the services or functions offered by the system
a) System requirements b) Functional requirements
c) Non-functional requirement d) None of them
93. ----- helps us to design the specific test scores that we need as it shows what inputs are required and what outputs are created
a) use case diagram b) Schema diagram c) Sequence diagram d) Activity diagram
94. ----- is best known and widely used of the agile methods.
a) Linear programming b) Extreme programming
c) Object oriented system development d) None of these
95. ----- occurs in real time systems use shared memory or message passing inter face
a) Timing Errors b) Programming errors c) Integration errors d) All of these
96. Which is the following is page fault?
a) Page fault occurs when a program accesses a page of another program
b) Page fault occurs when a program accesses a page in main memory
c) Page fault occurs when there is an error in particular page
d) Page fault occurs when a program access a page which is not present in main memory
97. Which of the following is a group of bits that tells the computer to perform a particular operation?
a) Accumulator b) Register c) Instruction code d) None of the above
98. Which of the following is performed by half adder?
a) Binary addition operation for 2 decimal inputs
b) Binary addition operation for 2 binary input
c) Decimal addition operation for 2 decimal input
d) Decimal addition operation for 2 binary inputs
99. In which of the following term the performance of cache memory is measured?
a) Chat Ratio b) Hit Ratio c) Copy Ratio d) Data Ratio
100. Content of the program counter is added to the address part of the instruction in order to obtain the effective address is called.
a) Relative address mode b) Index addressing mode
c) Register mode d) Implied mode

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